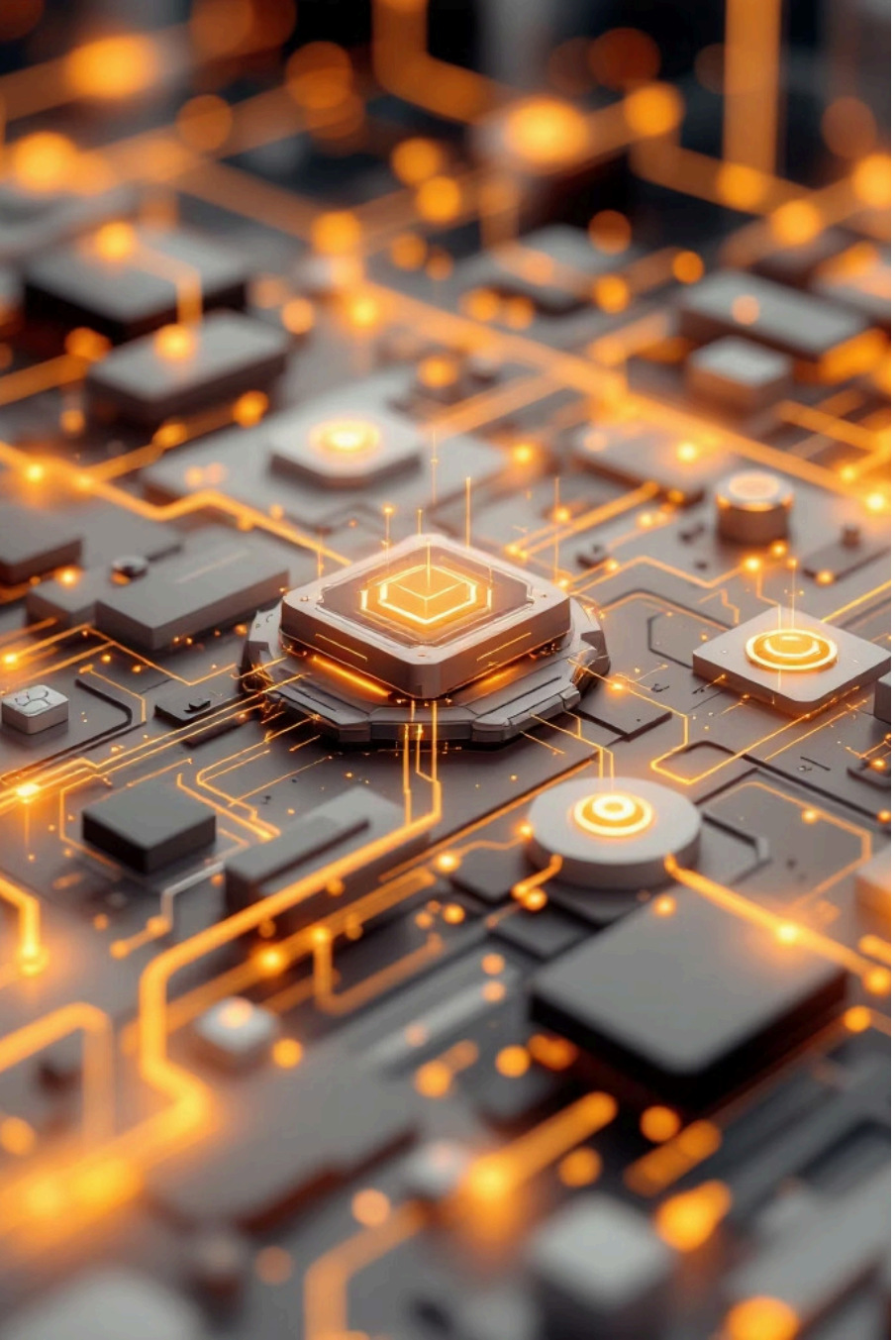


# Autonomous Content Engine (ACE)

## AI-Powered Enterprise Content Automation System


### Presenter

Soutik Mandal


### Hackathon

ET AI Hackathon 2026

### Theme

AI for Enterprise Content Operations

# Problem We're Solving

### Enterprise content creation is:

- ❌ Slow and manual
- ❌ Inconsistent across channels
- ❌ Prone to compliance risks

Multiple teams involved → delays + inefficiency

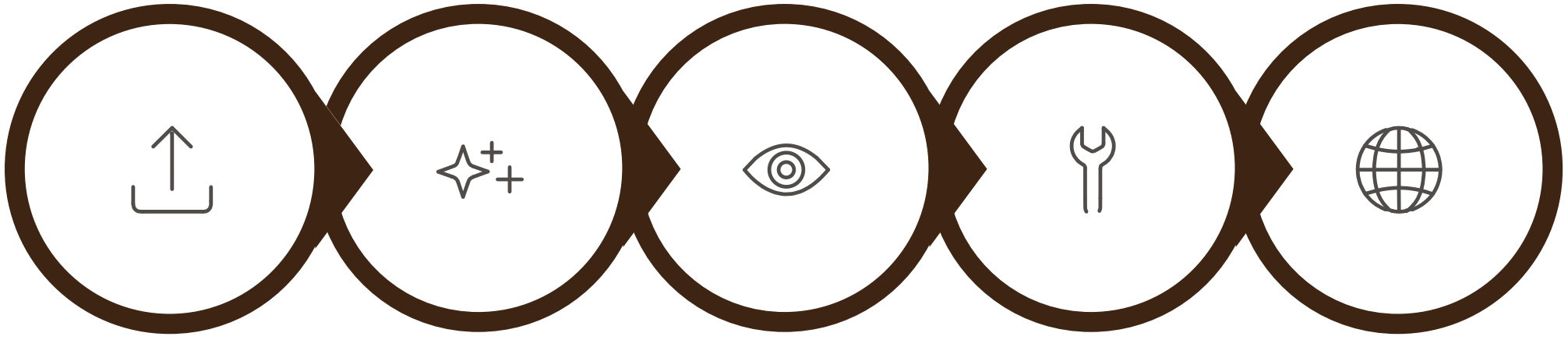
### No intelligent system for:

- End-to-end automation
- Real-time compliance
- Continuous improvement

👉 **Result:** High cost + slow turnaround + poor scalability

# Our Solution – ACE

A **multi-agent AI system** that automates the entire content lifecycle



Input Generate Review Fix Localize

The ACE pipeline takes raw inputs and transforms them through fully autonomous stages — from generation to publishing and continuous improvement.

### Fully Autonomous Workflow

### Multi-Agent Collaboration

### Self-Improving Feedback Loop

### Optional Human Approval

👉 **Outcome:** Faster, smarter, scalable content operations

# Multi-Agent Architecture

## Agents

✍️ Content Generator Agent

✅ Compliance & Brand Agent

🌍 Localization Agent

📢 Distribution Agent

📊 Analytics & Feedback Agent

## Orchestrator

- Manages workflow
- Handles failures & retries
- Maintains audit logs

### 👉 Why it's powerful:

Modular Design

Scalable & Extensible

Real Enterprise-Grade System

# How It Works

### Upload Input

PDF / text / product info

### AI Generates Content

### Compliance Agent Reviews & Corrects

### Content Localized

For multiple audiences

### Auto-Formatted for Platforms

### Simulated Publishing

### Analytics Agent Improves Future Outputs

👉 **Live UI shows:** Agent pipeline flow · Real-time decisions · Logs & metrics

# Impact & Innovation

**70–80%**

### ⏱️ Time Reduction

Reduction in content creation time

**↑**

### 📈 Consistency

Improved consistency across channels

**↓**

### ⚖️ Compliance Risk

Reduced compliance risks

**∞**

### 🔁 Optimization

Continuous optimization

## Innovation

### Full Lifecycle Automation

Not just generation — end-to-end

### True Multi-Agent Coordination

Agents collaborate in real time

### Knowledge → Content Transformation

Raw data becomes polished content

### Futuristic Real-Time Pipeline UI

Live visualization of every decision

# Why ACE Wins

## Core Strengths

- ✅ Solves a real enterprise problem
- ✅ Fully automated workflow
- ✅ Scalable across industries
- ✅ Strong UI + strong backend

## Future Scope

- Real API integrations (LinkedIn, CMS)
- Advanced analytics with real data
- Voice-based content generation
- Enterprise deployment

> 👉 **Vision:** "Reimagining how enterprises create and manage content with AI"